12 kW Wind Turbine for Iceland: Final Magnet Geometry (P1 generator)

Status: **Rev B, proposed for approval** · 2026-11-05 (Rev A, same day: special-size blocks 18.75 × 36.90 × 8.20; Rev B: catalogue 40 × 20 mm blocks, at Martin's request)
Owner: Category 4 (Generator) with 10 (Manufacturing and quality)
Builds on: 06 – Generator round 2 (frozen electromagnetic design), 14 – Nacelle prototype P1 (§4.5 magnets, §5.2 air gap budget), 18 – Generator manufacturing (§4.3 magnet purchase spec, §6.4 to §6.8 rotor build)
Analysis: `analysis/generator/magnets.py` → `magnets_summary.json` (2D nonlinear FEA of 11 magnet shapes, magnet eddy loss, demagnetisation, forces, tolerance scatter, step skew)
Geometry: `cad/generator_mfg/magnet_geom.py` (all magnet, seat and comb dimensions in this document come from here)
Pictures: `cad/generator_mfg/figures_magnets.py` → detail drawings M01 to M11 and isometric views MI01 to MI04; illustrated PDF in the project folder

Summary

Decision proposed: buy 40 × 20 mm flat magnets. Each pole is made of **four identical N40EH (or N40UH) blocks, 40 × 20 × 7.5 mm**, two side by side (20 across) and two along the stack (40 along), magnetised straight through the 7.5 thickness. The same block is used for north and south poles: it is simply turned over. **The rear row of blocks is turned 0.50° against the front row** (a step skew); this cancels the cogging that the wider pole would otherwise give. 160 + 8 identical blocks, one part number (NP1-220 Rev D).

Rev B changes only the magnet size, so that a **standard 40 × 20 face** can be ordered: makers stock 40 × 20 blocks and only the thickness is ground to order. Two consequences were designed in (§13): the poles become slightly wider (span 0.85 instead of 0.80), so the rows are skewed, and two 40 mm blocks plus the shim make 80.2 mm, so the magnets **overhang the 74 mm stator stack by 3.1 mm at each end** and the drum's magnet seat grows from 90 to 96 mm long.

	Doc 18 arcs (A)	Rev A blocks (F2t)	Rev B: 40 × 20 blocks (C2t)
Magnet per pole	4 arcs 7.20° × 18.35 × 7.70, radial	2 × 2 blocks 18.75 × 36.90 × 8.20	**2 × 2 blocks 20 × 40 × 7.50, rows skewed 0.50°**
Block size	special (curved)	special	**catalogue 40 × 20 face**, thickness ground to 7.50
Part numbers	2 (N and S)	1	**1** (NP1-220 Rev D), turned over for S
Pieces per generator	160 + 8	160 + 8	160 + 8
Piece mass / magnet mass	41 g / 6.65 kg	43 g / 6.92 kg	**46 g / 7.30 kg**
Pole span	0.80	0.80	0.85
Magnet length / stack	74 / 74	74 / 74	**80.2 / 74** (3.1 overhang each end)

	Doc 18 arcs (A)	Rev A blocks (F2t)	Rev B: 40 × 20 blocks (C2t)
Line EMF, 178 rpm, 100 °C, 28 turns	387 V	381 V	**380 V** (2D, overhang not counted, see §13)
Torque at the same current	803 Nm	790 Nm	787 Nm
Cogging, peak to peak	1.02 Nm	0.96 Nm	6.6 Nm without skew, **< 0.1 Nm with the 0.50° skew** (2D)
Torque ripple at load	1.75 %	1.56 %	2.8 % without skew, **1.97 %** with
Line EMF distortion (THD)	0.50 %	0.40 %	0.9 %
Magnet eddy loss, rated	10.9 W	12.8 W	**15.5 W** (budget 40 W)
Demag margin, worst short circuit, 120 °C, N40UH	27.7 %	32.2 %	**29.8 %**
Drum seat	Ø618.4 × 90 (doc 18)	Ø620.00 × 90	**Ø618.65 +0.05/0 × 96** (x 332 to 428)
Comb bars	40 × 82 long	40 × 82 long	**80 × 44 long** (one per row and gap)

Four findings change other documents:

1. **The open eddy-loss check is now done** (doc 14 open point 3, doc 18 M8). Magnet eddy loss at rated load is 11 to 13 W for any reasonable segmentation, against the 40 W rotor loss assumed in doc 06. **Splitting a pole into two blocks side by side cuts the loss by 3.6 ×** (it stops the largest current loops), so the flat design needs only **2 blocks along the stack** instead of 4 arcs. The magnet order no longer waits for anything.

2. **Doc 18 left out the bond line in the seat diameter.** The magnet seat Ø618.4 equals the magnet outer diameter, so the 0.15 mm glass-bead bond line would have pushed every magnet 0.15 mm into the sleeve space. The new seat Ø618.65 (Rev A: Ø620.00) includes the bond line.

3. **A flat block bonded to a round seat has a slightly thicker bond under its middle** (0.31 instead of 0.15 mm). This is fine for the adhesive and costs nothing measurable; one wide flat block per pole would have needed 0.73 mm and lost 6 % EMF, which is why it was rejected.

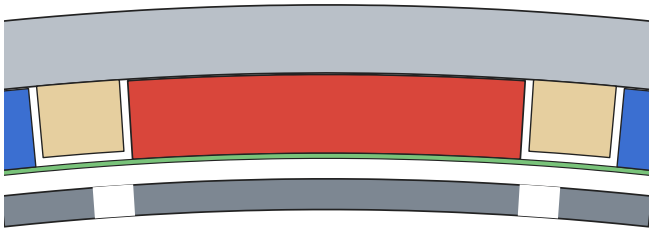
4. **(Rev B) A 0.85 pole span lands on the cogging peak of the 40/45 combination** (6.6 Nm), exactly as the span-0.84 cases did in Rev A. A step skew of half the 1° cogging period between the two block rows cancels it in the FEA (to < 0.1 Nm) and costs only 0.4 % EMF. This makes the 40 × 20 size usable; without the skew it would not be.

The flat-block route was chosen over the arcs because it is the **simpler and more controllable part to buy and inspect** (flat ground faces, micrometer-checkable, no grinding of radii, no radial-orientation tooling, one part number), it is **at least as good electrically**, and it removes the biggest mix-up risk in doc 18 (an N arc in an S position). Rev B goes one step further and uses a catalogue face size. Cost was not a criterion (project rule), but flat catalogue-size blocks are also the cheaper and faster part to source.

M01 Four ways to make one magnet pole: curved arc or flat blocks

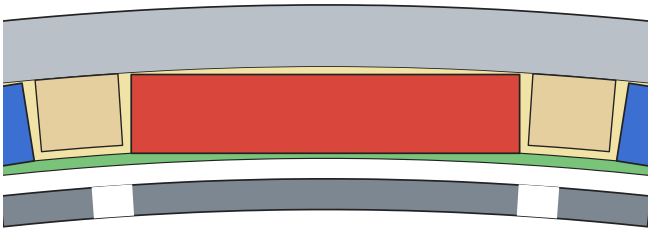
One pole cut across, drawn 9× real size. Drum on top, stator at the bottom, red = N pole, blue = next (S) pole.

A Curved arc 7.7 (doc 18)



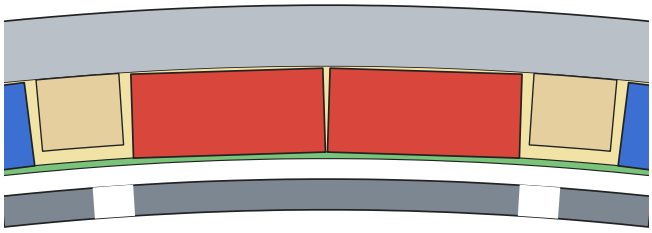
EMF 387 V torque 803 Nm cogging 1.01 Nm
magnet heat 11 W with 4 pieces along · magnets 6.65 kg
2 part numbers (N and S), ground curved faces

F1 One flat block 7.7



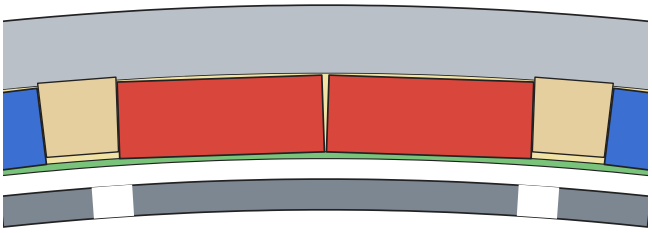
EMF 364 V torque 755 Nm cogging 1.06 Nm
magnet heat 9 W with 4 pieces along · magnets 6.57 kg
bond 0.73 mm thick in the middle, −6 % EMF

F2t Two 18.75 blocks 8.2 (Rev A)



EMF 381 V torque 790 Nm cogging 0.96 Nm
magnet heat 13 W with 2 pieces along · magnets 6.92 kg
Rev A: 18.75 × 36.90 × 8.20, special size

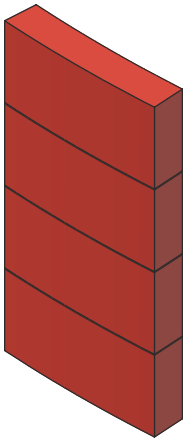
C2t Two 40 × 20 blocks 7.5 chosen



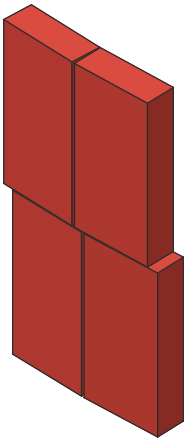
EMF 380 V torque 790 Nm cogging 6.6 → 0.05 Nm skewed
magnet heat 16 W with 2 pieces along · magnets 7.30 kg
catalogue 40 × 20 face, rows turned 0.5° (skew)

MI04 Curved arc pole (before) and flat block pole (now), isometric

Arc: 7.2° of the 9° pole pitch, 74 mm long. Blocks: 7.64°, 80.2 mm long, rows turned 0.5°.



Before: 4 curved arcs per pole,
2 part numbers (N, S)



Now: 2 × 2 blocks 40 × 20 per pole,
1 part number, turned over for S

1. What was fixed before and what was open

From docs 06, 14 and 18: 40 poles, 45 slots, outer rotor, stator tip radius 299.0 mm, mechanical gap 2.0 mm to a 0.5 mm glass-epoxy sleeve (moulded on a Ø602.0 mandrel, doc 18 §6.7), N40UH magnets 7.70 mm thick, pole span 0.80 of the 9° pole pitch (7.20°), stack 74 mm, 40 G11 comb bars between the poles, room-temperature toughened epoxy with 0.15 mm glass beads.

Open before this document:

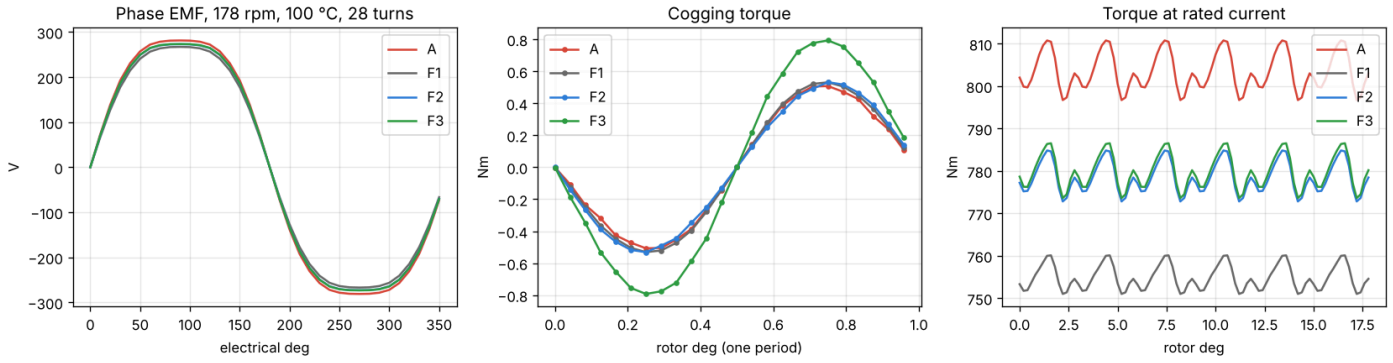
#	Open point	Closed by
1	Magnet axial segmentation not confirmed by an eddy-loss calculation (doc 14 open point 3, doc 18 M8)	§3: eddy loss computed for every shape and segmentation
2	Arcs need two part numbers (doc 18 M7), arc radii and radial orientation are the hardest features to buy and check	§2, §5: flat blocks, one part
3	Seat diameter did not include the bond line	§4: seat Ø618.65 (Rev A Ø620.00)
5	(Rev B) Use the catalogue 40 × 20 mm size	§13
4	Retention and tolerances never written down for the magnet itself	§6, §7

2. The options compared

Seven shapes were run in Rev A, and four 40 × 20 cases in Rev B (§13), through the same 2D nonlinear FEA (the doc 05/06 model, extended so that a magnet of any shape can sit on the fixed mesh). Every case uses the real P1 radii, the 0.15 mm bond line, the moulded sleeve bore r 301.0, a magnet inner face at r 301.5 on the pole (or block) centre line, the drum shell to r 323, the 74 mm stack, magnets at 100 °C and the same stator current (the doc 06 sizing current, 22.6 A rms with 28 turns).

Key	Shape	Seat Ø	Magnets	Line EMF	Torque	Cogging p-p	Ripple	Line THD	Eddy loss, 2 / 4 pieces along	Demag margin
A	Arc 7.20°, 7.70, radial (doc 18)	618.70	6.65 kg	387.4 V	803 Nm	1.02 Nm	1.75 %	0.50 %	27.4 / 10.9 W	27.7 %
F1	1 flat block 37.94 wide, 7.70	619.86	6.57 kg	364.3 V	755 Nm	1.06 Nm	1.21 %	0.25 %	22.7 / 9.0 W	28.5 %
F2	2 flat blocks 18.75 wide, 7.70	618.98	6.50 kg	375.5 V	779 Nm	1.06 Nm	1.54 %	0.36 %	13.2 / 8.0 W	31.3 %
F3	3 flat blocks 12.43 wide, 7.70	618.83	6.46 kg	376.2 V	780 Nm	1.58 Nm	1.66 %	0.44 %	7.8 / 5.7 W	30.2 %
F2w	2 blocks 19.70 wide (span 0.84), 7.70	619.01	6.82 kg	381.7 V	792 Nm	**5.40 Nm**	2.40 %	0.80 %	14.3 / 8.4 W	30.5 %
F2t	2 blocks 18.75 wide, 8.20 (Rev A)	619.98	6.92 kg	380.8 V	790 Nm	0.96 Nm	1.56 %	0.40 %	12.8 / 7.7 W	32.2 %
F2wt	2 blocks 19.70 wide, 8.00	619.61	7.09 kg	384.9 V	799 Nm	**5.53 Nm**	2.40 %	0.82 %	14.0 / 8.3 W	31.0 %

Key	Shape	Seat Ø	Magnets	Line EMF	Torque	Cogging p-p	Ripple	Line THD	Eddy loss, 2 / 4 pieces along	Demag margin
C1	1 block 40 wide, 4 × 20 along, 8.20	620.99	7.98 kg	374.9 V	778 Nm	2.73 Nm	2.00 %	0.67 %	10.0 W (4 × 20)	28.7 %
C2	2 blocks 20 wide, 2 × 40 along, 8.20	620.02	7.98 kg	388.7 V	807 Nm	**6.89 Nm**	2.80 %	0.97 %	14.8 W (2 × 40)	31.2 %
C2t	**2 blocks 20 wide, 2 × 40 along, 7.50**	**618.62**	**7.30 kg**	**381.1 V**	**791 Nm**	**6.64 Nm** (skewed < 0.1)	2.83 % (skewed 1.97 %)	0.92 %	**15.5 W (2 × 40)**	**29.8 %**
C2s	2 blocks 20 wide, 2 × 40 along, 7.00	617.62	6.81 kg	374.8 V	777 Nm	6.41 Nm	2.83 %	0.89 %	16.0 W (2 × 40)	28.7 %



What the table says:

- **Flat blocks lose EMF because they hold less magnet, not because they are flat.** With 2 or 3 blocks the loss is the same 3 %: blocks that fit the pole at the inner face leave small wedge-shaped gaps towards the drum, so 2 to 3 % less magnet. One wide block (F1) loses 6 % because its edges stand 0.6 mm further from the stator.
- **Widening the pole is a trap.** Span 0.84 (F2w, F2wt) wins the EMF back but lands on a cogging peak of the 40/45 combination: 5.4 Nm instead of 1 Nm, and the ripple and EMF distortion double. Rev A therefore kept the doc 06 span of 0.80. Rev B accepts 0.85 (set by the 20 mm catalogue width) and removes the cogging peak with a step skew of the two rows (§13.4).
- **A 0.5 mm thicker block (F2t) wins the EMF back** without touching the span, the stator or the winding: 381 V at 28 turns (doc 06 target 380 V), cogging and ripple slightly lower than the arcs, best demagnetisation margin. It costs 0.27 kg more magnet, 0.8 mm on the seat diameter and 0.5 mm of drum yoke (13.0 mm left under the seat; the doc 06 FEA used 12 mm).

- **Three blocks (F3)** would cut the eddy loss further but raise cogging (1.6 Nm) and give 240 small pieces. No reason to go there.
- **Demagnetisation.** Worst case is the doc 06 transient short circuit (2 × the steady short-circuit current, d-axis, magnets at 120 °C). All flat shapes are better than the arcs. Note: in the first run the flat shapes showed a false demagnetising peak in mesh elements cut by the slanted block sides; the check now uses only elements fully inside a magnet (as for the arcs, whose edges follow the mesh).

Rev A chose F2t. **Rev B chooses C2t** (40 × 20 × 7.50, rows skewed 0.50°, §13); the rest of this document defines C2t. The C cases are computed on the 74 mm stack; the 3.1 mm overhang at each end is not in the 2D model.

3. Magnet eddy loss and segmentation (closes doc 14 point 3, doc 18 M8)

Method: time-stepped FEA at rated load over one electrical period (90 steps; the stator-frame field repeats every 18° of rotation, so 4 periods of the rotor-frame history are built from it exactly), field history followed in every magnet node in the rotor frame, eddy current density $\sigma \cdot (-\partial A/\partial t)$ with **zero net current in each piece**, resistivity 1.5 μΩ m. The 2D result is for infinitely long pieces; the finite length of each piece is applied with the exact factor for a rectangle in a uniform alternating field (the Prandtl torsion analogy). The dominant field frequencies seen by the magnets are 44.5, 89, 133.5 and 178 Hz: the 40/45 winding's space harmonics, not slot ripple, so the skin depth (> 25 mm) is far larger than the magnet and the resistance-limited method is valid.

Pieces along the stack	1	2	3	4	6
A, arc 38 mm wide	45.3 W	27.4 W	16.7 W	10.9 W	5.5 W
F2t, block 18.75 wide	15.8 W	**12.8 W**	10.0 W	7.7 W	4.7 W
F3, block 12.4 wide	8.8 W	7.8 W	6.7 W	5.7 W	4.1 W
C2t, block 20 wide (on 74 mm)	18.7 W	14.9 W	11.5 W	8.7 W	5.2 W

Result: **2 blocks along the stack**. For Rev B (two 40 mm blocks, the whole 80.2 mm in the field) the loss is **15.5 W** at rated speed and about 30 W at the 250 rpm design overspeed (loss ∝ speed²); Rev A had 12.8 W. Doc 06 assumed 40 W rotor loss; the magnets sit on the drum, which is cooled by the outside air, so their temperature rise from this is a few kelvin. Going to 4 blocks along would save 5 W for twice the pieces and three shim gaps instead of one, and is not worth it.

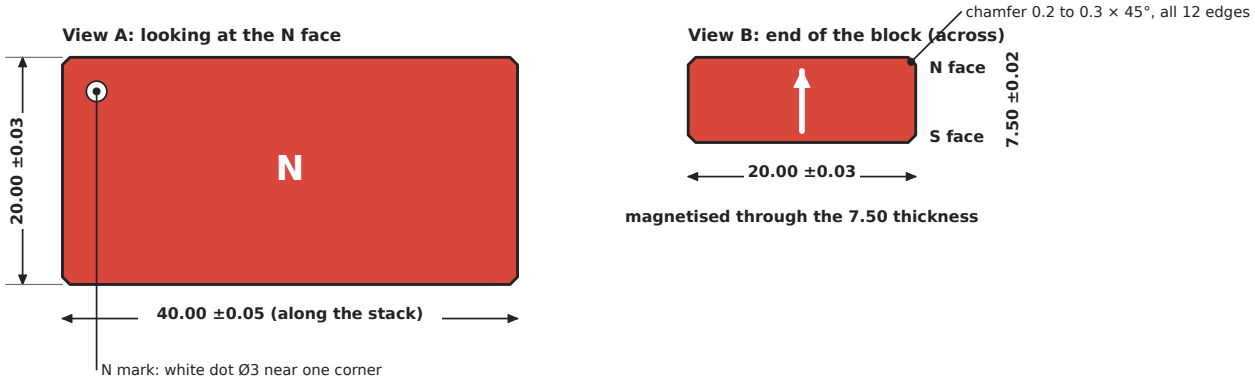
PWM ripple from the SiC active rectifier (doc 12) adds little: with Ld ≈ 14 mH the switching ripple current is well under 1 A, against 23 A rated. It is checked on the bench by the magnet temperature probe (doc 15).

4. Final geometry

4.1 The block NP1-220 Rev D

M03 Detail drawing NP1-220 Rev D Magnet block 40 × 20 × 7.5 (bought part)

Drawn 10× real size. Dimensions in mm. Tolerances as marked, all others ±0.05.



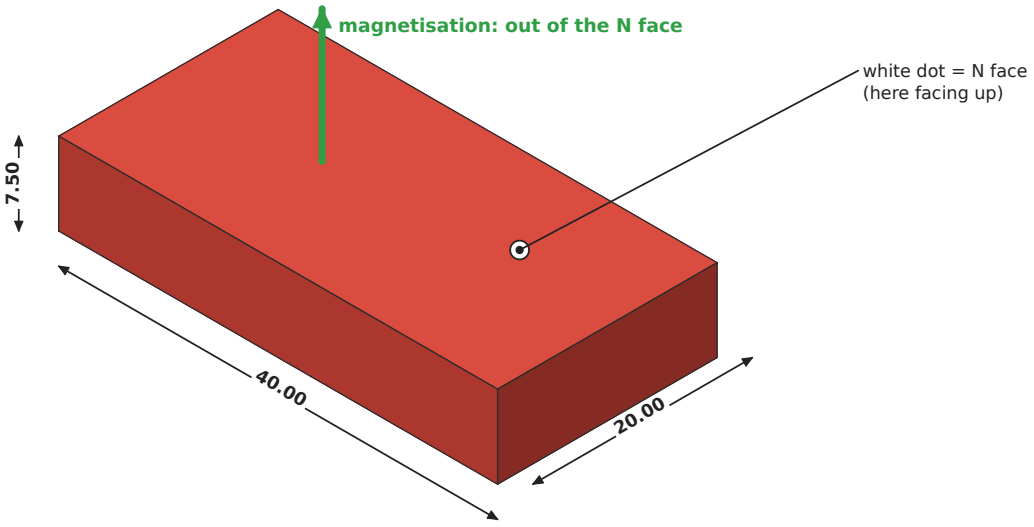
Flatness of N and S faces 0.02
Parallelism N to S face 0.02
Squareness of the sides to the N face 0.05
Magnetic axis within 2° of the face normal

Part	NP1-220 Rev D magnet block, catalogue 40 × 20 face
Material	Sintered NdFeB N40EH preferred, N40UH accepted (doc 19 §12)
Magnetised	Fully, parallel, through the 7.50 thickness
Coating	NiCuNi ≥ 15 µm + epoxy ≥ 15 µm (total 30 to 45 µm, in the sizes)
Mark	White dot Ø3 on the N face, one corner
Mass	45.6 g each
Quantity	160 + 8 spare + 8 test pieces per generator
Replaces	Rev C 18.75 × 36.90 × 8.20 (doc 19 Rev A), arcs -N/-S
Moment	Each piece within ±1.5 % of the lot mean, value per serial
Surfaces	Faces ground, Ra ≤ 1.6 before coating
Scale	10 : 1 on this sheet

Doc 19 Rev B · Wind turbine 12 kW · P1 generator

MI01 The magnet block NP1-220 Rev D in 3D (isometric, 8× real size)

One part for every pole. The arrow shows the magnetisation, from the S face to the N face.



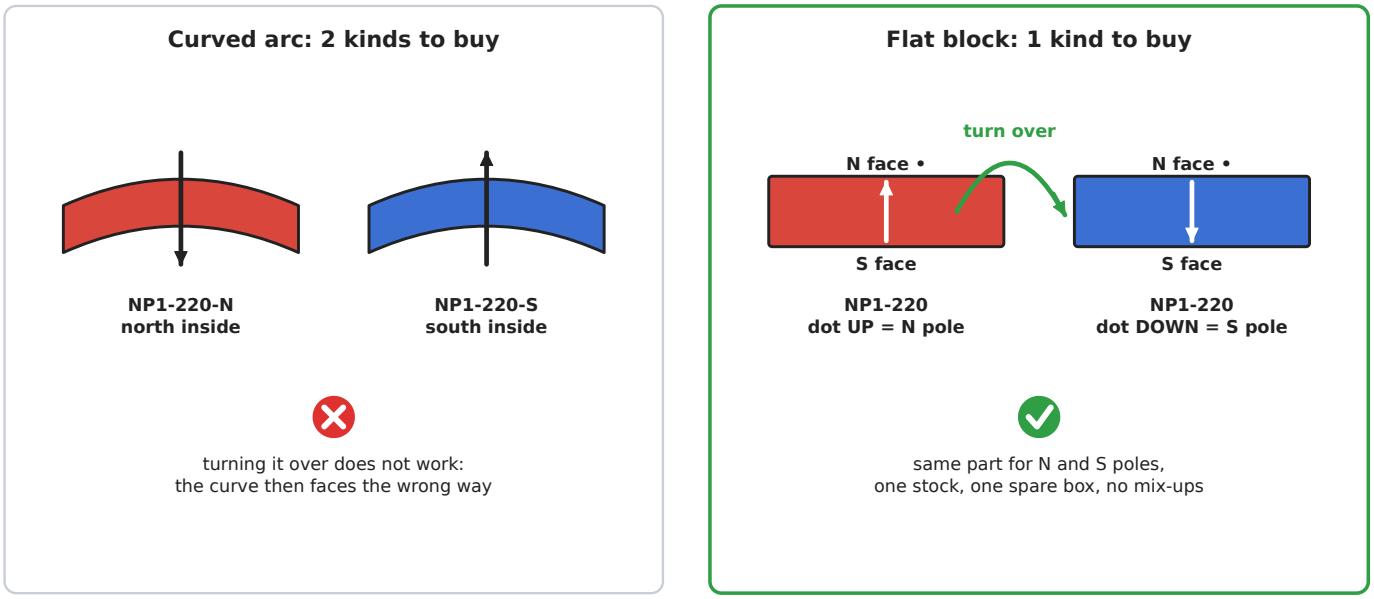
N40EH or N40UH, 46 g, 40 × 20 face like a small domino. Pulls ≈ 280 N (28 kg) on thick steel.
Flat faces ground, edges chamfered 0.2 to 0.3, NiCuNi + epoxy coated, white dot on the N face.
160 per generator (+8 spare): 40 poles × 2 across × 2 along.

Feature	Value
Size	**20.00 ±0.03 (across) × 40.00 ±0.05 (along) × 7.50 ±0.02 (thickness)**; catalogue 40 × 20 face
Edges	Chamfer 0.2 to 0.3 × 45° on all 12 edges
Faces	N and S faces ground, flatness 0.02, parallelism 0.02; sides square to the N face within 0.05
Magnetisation	Parallel, through the 7.50 thickness, fully saturated; magnetic axis within 2° of the face normal
Mass	45.6 g
Count	160 per generator + 8 spare + 8 test pieces (unmagnetised acceptable) for Q1
Marking	White dot Ø3 on the N face near one corner

4.2 One part, two poles

M02 Why flat is easier: ONE magnet part, turned over for the other pole

A flat block is magnetised straight through its thickness. Turn it over and the north face points the other way.



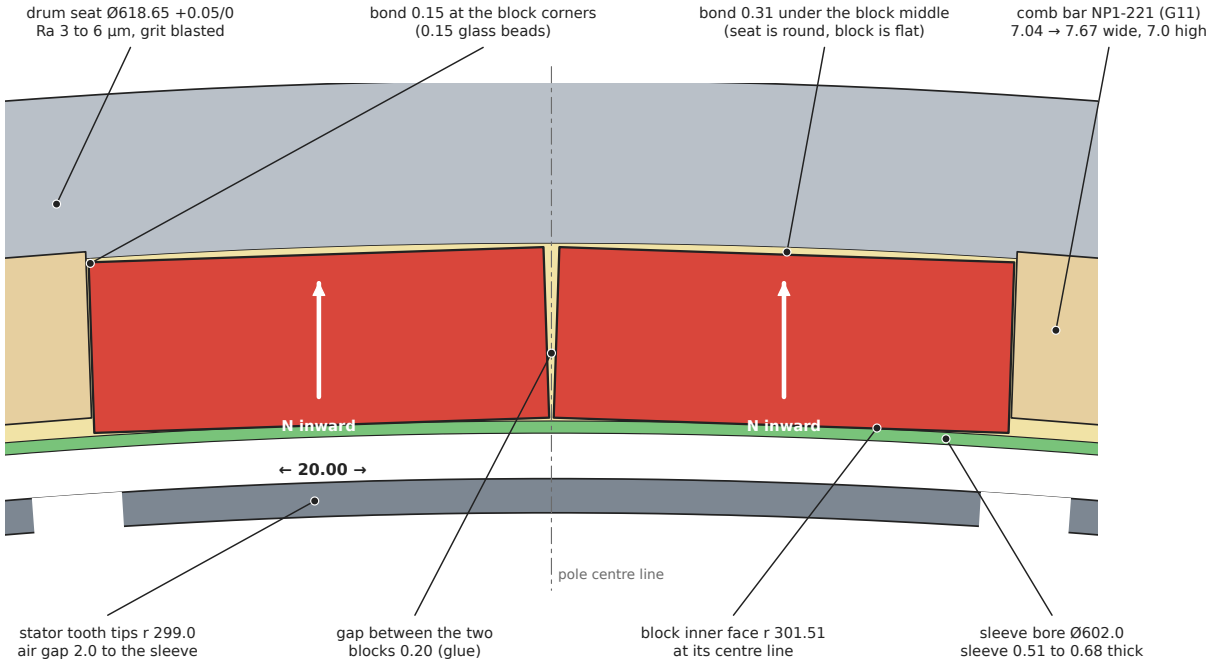
Arrow = direction of magnetisation (from S face to N face, inside the magnet). The white dot marks the N face.

At an N pole the block goes in **dot up** (N face towards the stator); at an S pole the same block goes in **dot down** (N face against the drum). This removes doc 18 M7 completely: one part number, one stock, one spare box.

4.3 The pole on the drum

M04 Detail: one pole of two blocks on the drum seat (cross-section, 20× real size)

Everything a drawing checker needs: radii from the generator axis, bond line, sleeve, comb bars.



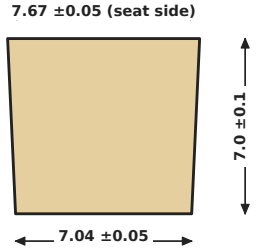
Pole pitch 9.000°, pole span 7.64°, block centres at ±1.92°
Blocks: 20.00 × 40.00 × 7.50, magnetised through 7.50
Rows: front −0.25°, rear +0.25° from the drawn centre (skew 0.50°)
Drum yoke under the seat: 13.7 mm (shell OD Ø646)

Feature	Value (radii from the generator axis)
Pole pitch / span	9.000° / 7.64° (0.85); block centre lines at ±1.92° from the pole centre
Step skew	**Front row pole centres −0.25°, rear row +0.25°** from the nominal pole centre (0.50° = 2.7 mm at the seat)
Drum seat	**Ø618.65 +0.05/0**, **x 332 to 428** (96 long; was x 335 to 425), run-out ≤ 0.02 to the cup seats, Ra 3 to 6 µm after grit blasting
Bond line	0.15 at the block corners (0.15 mm glass beads), 0.31 under the block middle
Block inner face	r 301.51 at the block centre line, r 301.68 at the block edges
Sleeve	Moulded on the Ø602.0 mandrel: 0.51 thick at the block centres, 0.68 at the block edges; sleeve length x 332 to 428
Gap between the two blocks of a pole	0.20 at the inner corners, widening to 0.70 at the outer corners (glue)
Comb bars NP1-221	G11, trapezoid 7.04 (inner) to 7.67 (seat side) × 7.0 high × **44.0 long, two per pole gap** (front bar x 336 to 380, rear bar x 380 to 424, rear bar turned 0.50°), faces at 2.58° to the bar centre line so they lie flat against the neighbouring blocks; 80 + 8 spare
End rings NP1-222	G11, OD Ø618.65 −0.1/−0.2, ID Ø604.0, **7.9 wide**, 40 notches for the comb bar ends; the rear ring is clocked 0.50° against the front ring
Axial layout	Block rows **x 339.9 to 379.9 and x 380.1 to 420.1** (stack x 343 to 417: 3.1 mm overhang at each end), 0.2 mm PTFE shim between rows (removed after cure)
Drum yoke under the seat	13.7 mm (shell OD Ø646 unchanged; the Ø646 band grows to x 332 to 428 with the seat)

M05 Detail drawings NP1-221 comb bar (80 + 8 spare) and NP1-222 end rings (2)

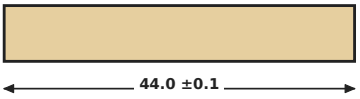
Glass-epoxy G11 (class H). The comb bars set where every pole sits. Dimensions in mm.

Section (22× real size)



faces 2.58° to the bar centre line (each side)
wide face is glued to the drum seat

Length (2× real size)



front bar x 336 to 380, rear bar x 380 to 424
ends square, edges broken 0.3

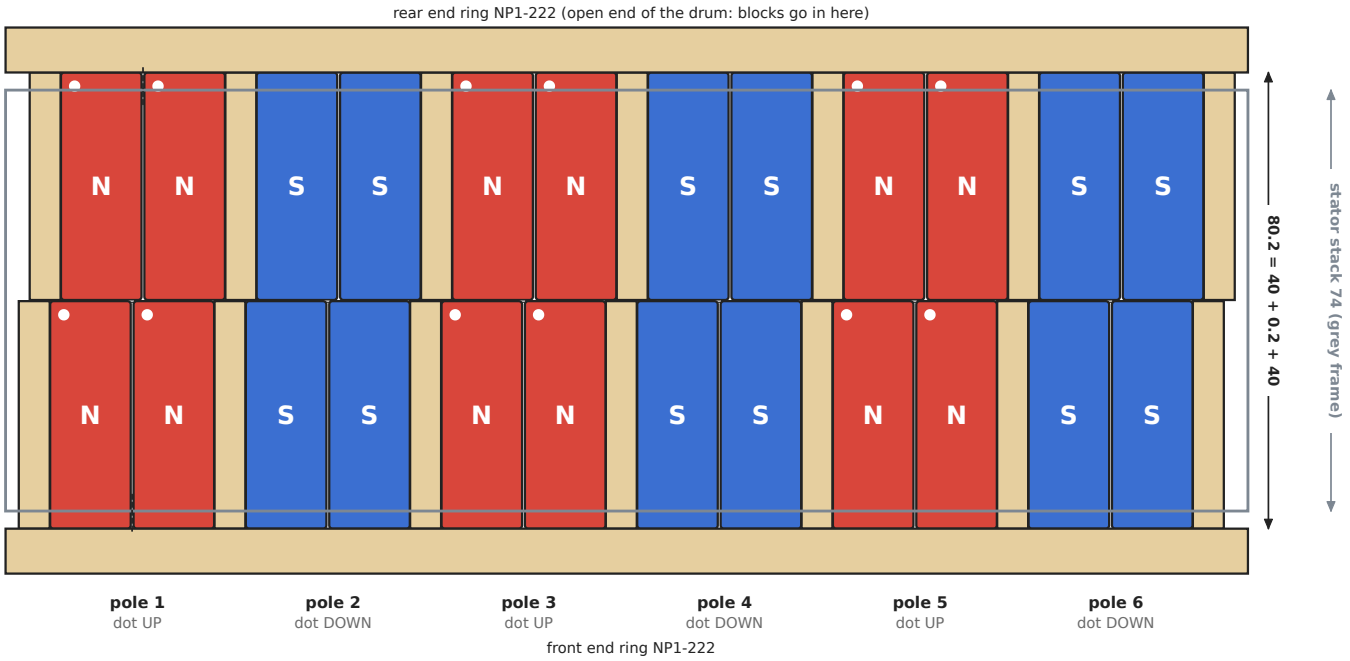
NP1-222 end ring (front and rear, same part)

G11 ring, OD Ø618.65 −0.1/−0.2 (glued in the seat),
ID Ø604.0 ±0.1 (stays 1.0 mm outside the sleeve bore),
width 7.9 −0.1 (x 332 to 339.9 front, x 420.1 to 428 rear),
40 notches 7.1 deep × comb bar width +0.2, pitch 9.000°.

Two bars per pole gap: one for the front row, one for the rear row, ends in the end-ring notches. Rear ring and rear bars are turned 0.50° against the front ones (skew). Notch 1 from hub bolt hole 1.

M06 Where every block goes: the drum seat unrolled, seen from the stator

Part of the ring, 6 of the 40 poles. Each pole = 2 blocks across × 2 blocks along = 4 blocks of the SAME part.

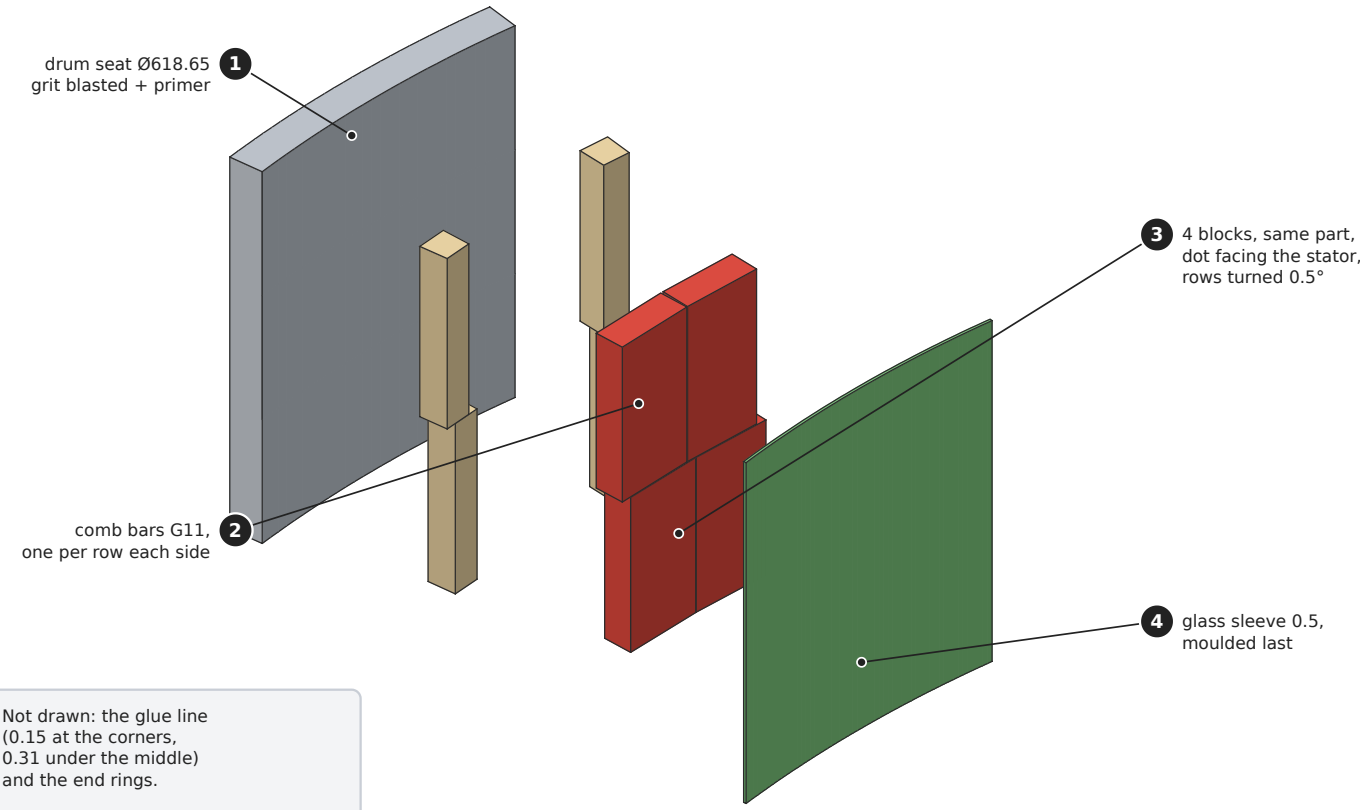


tan = G11 comb bars and end rings
white strip = 0.2 mm PTFE shim (taken out after cure)

- rear row turned 0.50° (2.7 mm) against the front row
- blocks overhang the stack 3.1 mm at each end
- white dot visible = N pole (dot faces the stator)
- N and S always alternate, 20 + 20 poles

MI02 One pole pulled apart (isometric), seen from the stator side

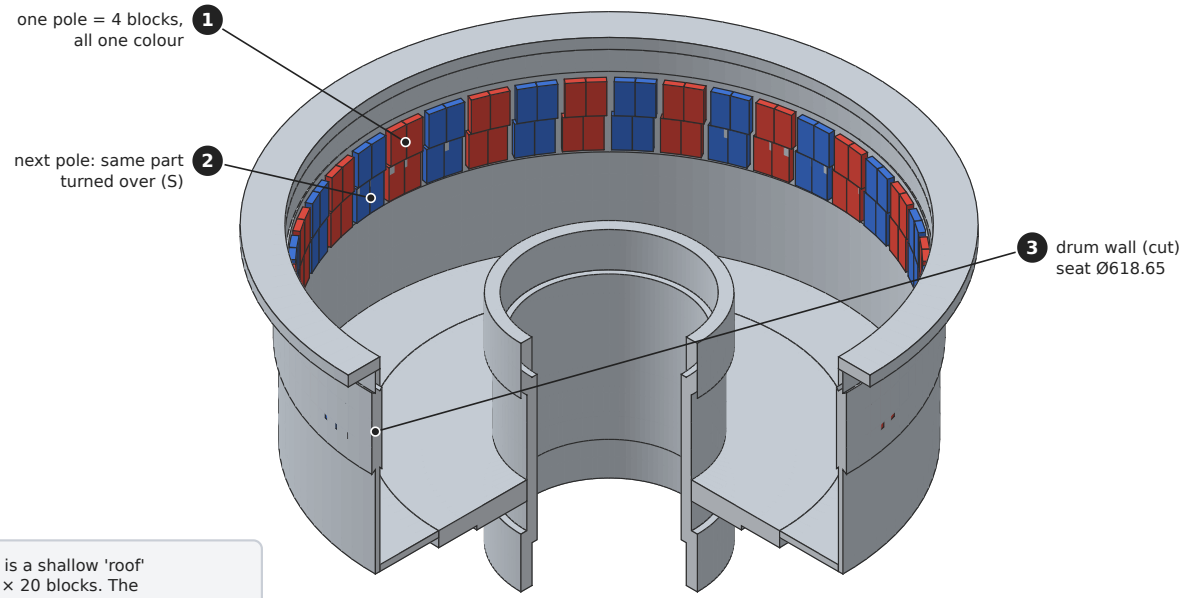
From the back: drum seat, comb bars, four blocks, glass sleeve. Each layer is pulled towards you (towards the axis).



Not drawn: the glue line (0.15 at the corners, 0.31 under the middle) and the end rings.

MI03 The rotor drum with its 160 flat blocks (isometric, cut open)

Open end up, as on the jig. A quarter is cut away. 40 poles: red N, blue S, each 2 × 2 blocks.



Each pole is a shallow 'roof' of two 40 × 20 blocks. The rear row is turned 0.5° so the cogging of the two rows cancels.

The two blocks of one pole **push each other apart** (same polarity side by side), so each one settles against its own comb bar by itself. The comb bars therefore set the pole position, and the 0.20 mm centre gap fills with glue. Because the front and rear rows have their own comb bars, the 0.50° skew is set simply by where the bars sit: the front ring and bars on the nominal grid −0.25°, the rear ones +0.25°, both from the same indexing template (two marked positions).

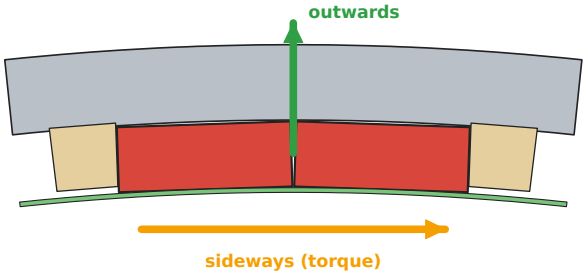
5. Flat or arc: buying and making

Aspect	Arc segments (doc 18)	Flat 40 × 20 blocks (Rev B)
How the maker produces it	Pressed in a radially oriented die (tooling per size) or cut from blocks with much waste; both radii ground with profiled wheels	Sliced from standard pressed and sintered blocks; the 40 × 20 face is a stock size, only the 7.50 thickness is ground to order; no tooling
Tolerances commonly held	Thickness ±0.05, radii ±0.05, arc angle ±0.1°	±0.02 to ±0.05 on every size, flatness 0.02
Magnetisation	“Radial” for a 7.2° segment is in practice parallel along the centre line; needs a shaped fixture	Parallel through thickness, the standard case for every maker
Polarity variants	2 part numbers (curve fits one way only)	1 part number
Coating	Even coat on curved faces is harder; thin arc edges chip	Even coat on flat faces, chamfered edges
Incoming inspection	Radius gauges or CMM	Micrometer and go/no-go gauges
Typical lead time (sintered NdFeB, custom size)	6 to 10 weeks incl. tooling and first article	3 to 6 weeks after drawing approval
Number of qualified makers	Fewer (radial tooling)	Many
Effect on the machine	Reference	−2 % EMF at the same turns (chosen to hit the 380 V target); cogging and ripple as low as the arcs only with the 0.50° row skew

6. Retention: what holds the magnets

M08 The forces on one pole of magnets, and what carries them

Outer rotor: the magnets sit inside the drum, so spinning pushes them OUTWARDS into the steel.



Load on one pole (4 blocks)	size	carried by
Spinning at 250 rpm (overspeed)	**38 N outwards**	drum steel
Magnetic pull, no load	**37 N outwards**	drum steel
Magnetic pull, rated load	**131 N outwards**	drum steel
Torque share, rated (local peak)	**85 N sideways**	glue shear, comb bar
Short circuit, 4 × torque	**≈ 264 N sideways**	glue shear, comb bar
Cold −40 °C (steel shrinks more)	**≈ 0.02 mm at corners**	tough room-cure glue
If a block ever cracks	**pieces stay put**	sleeve + end rings

Glue area per pole 3200 mm². Rated shear 26 kPa, short circuit ≈ 82 kPa, against ≥ 5 MPa at 120 °C: more than 50× margin. The glue never carries a pull-off load, so a block cannot fall into the air gap even with no glue.

Load (one pole, 4 blocks)	Value (from FEA unless noted)	Carried by
Centrifugal at 250 rpm	38 N outwards (hand calculation)	Drum steel (outer rotor)
Net magnetic force, no load	37 N outwards	Drum steel
Net magnetic force, rated load	131 N outwards	Drum steel
Tangential, rated load (local peak on one pole)	85 N	Glue shear and comb bars
Tangential, short circuit (4 × torque)	≈ 260 N	Glue shear and comb bars
Differential shrinkage steel/NdFeB, 20 to −40 °C	≈ 0.02 mm at the block corners	Toughened room-cure glue (qualified on Q1)

Glue area per pole is 3,200 mm², so the rated shear is 27 kPa and the short circuit case about 83 kPa, against ≥ 5 MPa adhesive strength at 120 °C: more than 50 × margin. Every radial load pushes the blocks into the drum, so the glue never carries a pull-off load. Three independent things keep a block out of the air gap: the glue, the comb bars and end rings, and the moulded glass sleeve over all magnets (it also holds the pieces if a block ever cracks). The thermal-mismatch movement at the block corners is set by the half-diagonal of the block (22.4 mm), almost the same as the doc 18 arc (21.5 mm), so the doc 18 adhesive choice and the Q1 cold-cycling test stand unchanged; Q1 must use real NP1-220 Rev D blocks. The 3.1 mm of block that overhangs the stack at each end still sits fully on the seat and under the sleeve.

7. Tolerances

M07 From the drum to the stator: every layer and its tolerance (radial, mm)

The sleeve is moulded on the Ø602.0 mandrel, so magnet and bond scatter only change the sleeve thickness, not the air gap.



	r 309.325 +0.025/0 (Ø +0.05/0)
0.15	+0.05 / 0 (beads set the minimum)
7.50	±0.02
0.51	result: 0.49 to 0.61
	r 301.00 +0.025/−0.01, run-out ≤ 0.03
2.00	≥ 1.80 anywhere (doc 18 §7)
	r 299.00

Magnet inner face at a block centre:
nominal r 301.513
lowest r 301.49 (thick block, small seat)
highest r 301.61 (thin block, big seat, thick bond)

Sleeve 0.49 to 0.61 at the block centre, up to 0.78 at the block edges: inside the 0.4 to 0.8 range tested on Q1.

Air gap: unchanged, set by the mandrel.

Reject a drum if the seat is smaller than Ø618.65: the blocks would push into the sleeve space. Larger seat: just thicker sleeve.

Radial. The sleeve is moulded on the mandrel, so the magnet and bond scatter only change the sleeve thickness, not the air gap (doc 18 M6). With seat +0.05/0 on the diameter, bond +0.05/0 and block ±0.02, the inner face at a block centre lies between r 301.49 and r 301.61: the sleeve is 0.49 to 0.61 at the block centres and up to 0.78 at the block edges, inside the 0.4 to 0.8 range to be qualified on Q1. A drum with a seat smaller than Ø618.65 is rejected. The doc 14 air gap budget item “magnet thickness and bond line 0.050” drops out of the gap budget (it is replaced by the mandrel and sleeve bore tolerance already in doc 18 §6.8).

Across and along. Block width ±0.03 gives a centre gap between the two blocks of 0.14 to 0.26 mm (never touching, glue fills it). The comb bar width ±0.05 and the 0.1 mm glue clearance per side keep the pole span within ±0.02°. Block length ±0.05 with the 0.2 mm shim keeps the rows within 80.2 ±0.15, so the overhang is 3.1 ±0.2 at each end; the stator stack tolerance no longer matters because the magnets are longer than the stack. **The skew angle matters:** in the FEA a mean skew of 0.45° or 0.55° leaves 1.0 Nm, 0.40° or 0.60° leaves 2.0 Nm, 0.50° leaves < 0.1 Nm. The two template positions (front and rear) are therefore machined 0.50 ±0.02° apart (±0.11 mm at the seat), which keeps it below about 0.5 Nm. Random errors of single poles average out over the 40 poles and are covered by the scatter study below.

Magnetic. Tolerance study (pole remanence and position scatter, cogging over a full slot pitch; the 72° FEA section repeats the scatter pattern five times, so this is an indication, not a statistical result). For Rev B the same scatter is applied to both rows of a pole:

Case	Rev A (F2t)	Rev B (C2t), no skew	Rev B, rows skewed 0.50°
Nominal	0.96 Nm	6.64 Nm	**0.02 Nm**

Case	Rev A (F2t)	Rev B (C2t), no skew	Rev B, rows skewed 0.50°
Poles sorted to ±0.5 %, strong next to weak (doc 18 sorting rule)	1.14 Nm	6.88 Nm	**0.24 Nm**
Unsorted, every pole random within ±1.5 %	2.74 Nm	8.03 Nm	1.37 Nm
One pole 3 % weak (e.g. a chipped block)	2.41 Nm	7.82 Nm	1.28 Nm
Pole centres random within ±0.05° (±0.26 mm at r 301)	2.35 Nm	7.70 Nm	**1.13 Nm**
Pole centres random within ±0.10° (doc 18 pole-map limit)	8.50 Nm	13.2 Nm	6.32 Nm
Pole centres random within ±0.25°	14.8 Nm	16.7 Nm	10.8 Nm

With the skew, Rev B is as good as or better than Rev A in every scatter case; without it, it is 3 to 7 times worse. The tolerance rules below apply unchanged.

Conclusions:

- **Remanence:** sorting matters. Unsorted pieces (±1.5 %) almost triple the cogging; the doc 18 sorting rule (poles within ±0.5 %, strong next to weak) keeps it at 1.1 Nm. Keep it.
- **Pole position matters much more than remanence.** At the doc 18 pole-map limit of ±0.1° the cogging rises to about 8.5 Nm (1.2 % of rated torque). The section model repeats the error pattern five times, which adds the errors coherently, so the real value over 40 independent poles is lower, but the trend is clear. **Tighten the comb bar placement to ±0.03° (±0.16 mm at the seat) and the pole-map acceptance to ±0.05°** (doc 18 §6.4 and §6.6). With trapezoid comb bars in notched end rings and blocks that push themselves against the bars, ±0.05° is reachable with the aluminium indexing template already planned.

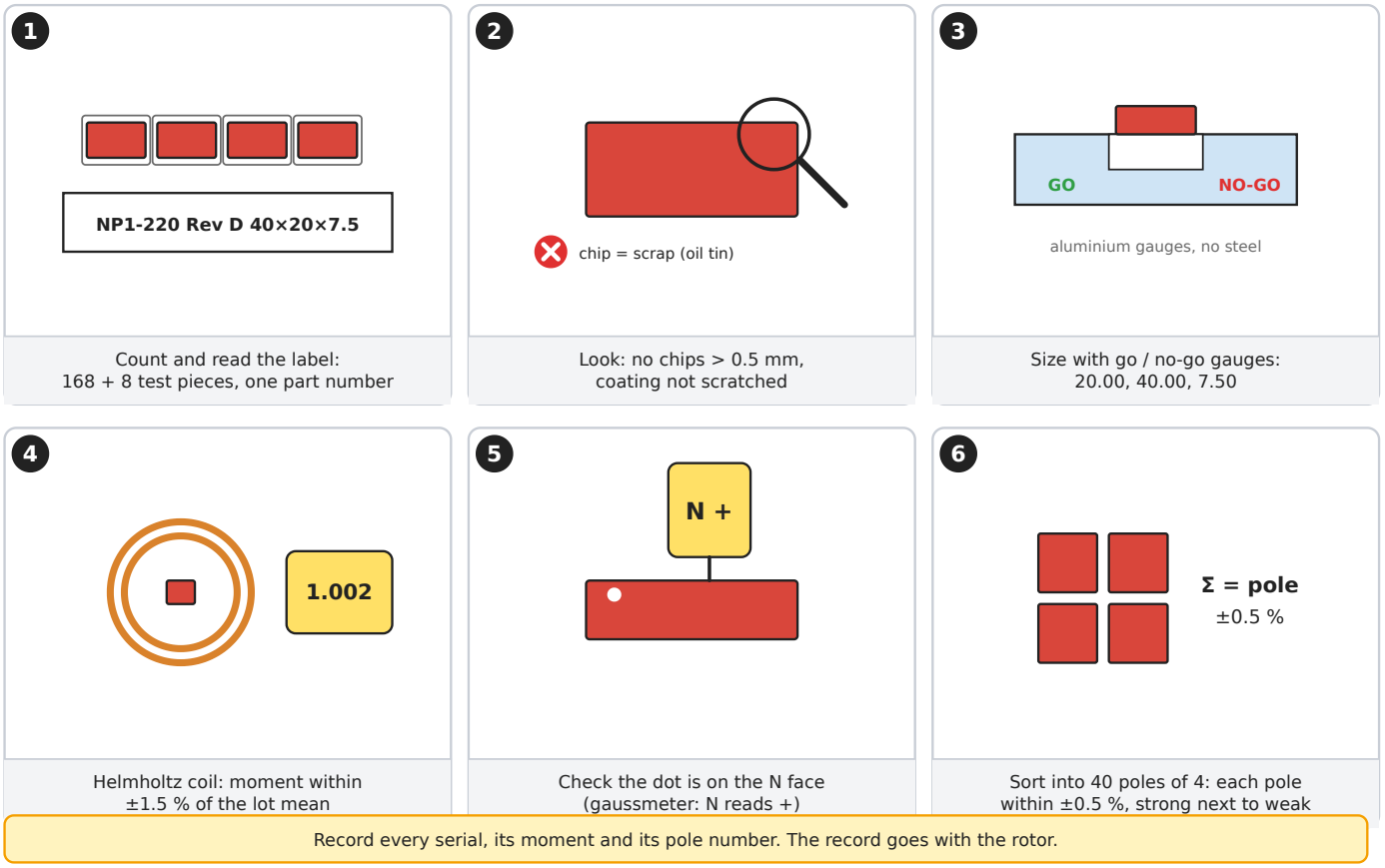
8. Purchase specification NP1-220 Rev D (send with drawing M03)

Item	Requirement
Part	NP1-220 Rev D magnet block 40 × 20 × 7.5, drawing M03 (doc 19 Rev B)
Material	Sintered NdFeB (GB/T 13560 designation), quote both: **N40EH preferred** (Br ≥ 1.25 T, HcJ ≥ 2,388 kA/m / 30 kOe) and **N40UH accepted** (Br ≥ 1.26 T, HcJ ≥ 1,990 kA/m / 25 kOe), values at 20 °C (§12). Grain-boundary-diffused Dy/Tb grades accepted if they meet all values
Export	State country of manufacture, origin of the Dy/Tb, and lead time including any export licence
Test data per lot	J-H and B-H demagnetisation curves at 20, 60, 100, 120 and 150 °C on samples from the lot; temperature coefficients of Br and HcJ; density
Size	40.00 ±0.05 × 20.00 ±0.03 × 7.50 ±0.02 mm **including coating**; chamfer 0.2 to 0.3 on all edges; flatness and parallelism of the pole faces 0.02. A stock 40 × 20 block ground to 7.50 is welcome; state if a stock thickness is offered instead (§13.4)
Magnetisation	Parallel through the 7.50 dimension, fully saturated; deviation of the magnetic axis from the face normal ≤ 2°
Uniformity	Magnetic moment (Helmholtz coil) of every piece measured and reported against its serial; all pieces within ±1.5 % of the lot mean
Coating	NiCuNi ≥ 15 µm + epoxy ≥ 15 µm; ≥ 500 h neutral salt spray (ISO 9227) without red rust; cross-cut adhesion class 0 to 1; adhesion of the chosen structural adhesive to the coating approved by the adhesive maker
Corrosion of the bare alloy	Low-weight-loss grade: PCT 121 °C, 2 bar, 100 % RH, 96 h, weight loss ≤ 5 mg/cm²
Marking	White dot Ø3 on the N face, one corner; lot number on every box
Quantity	168 magnetised + 8 test pieces (unmagnetised acceptable) per generator
Packaging	Every block in its own non-magnetic spacer, alternating (repelling) layout, steel-shielded box meeting the IATA PI 953 field limit for air freight
First article	5 pieces measured (size, moment, coating thickness, one demagnetisation curve) and approved before the lot is released
Documents	Material certificate, per-piece moment list, coating report, salt-spray and PCT reports, RoHS/REACH declaration

9. Incoming inspection and bonding

M09 When the box arrives: 6 checks for every block (in the magnet room)

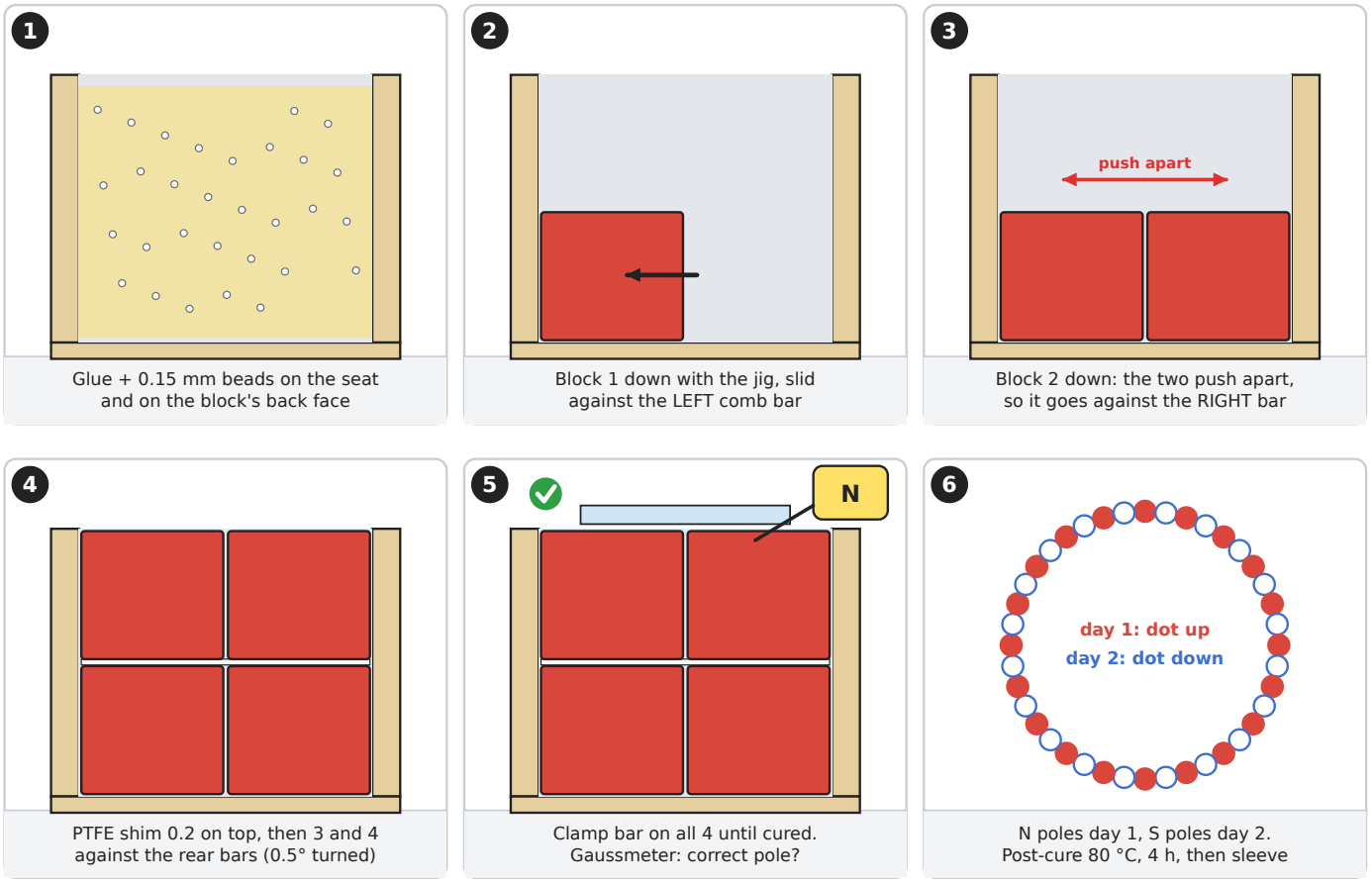
Two people. Only one block out of its spacer at a time. Gloves and glasses.



Incoming inspection replaces doc 18 §6.5 “receive and sort”: count, look (no chips > 0.5 mm), size with aluminium go/no-go gauges, moment in the Helmholtz coil (±1.5 % of the lot mean, within ±1 % of the supplier’s value), check the dot is on the N face with the gaussmeter, then **sort into 40 poles of 4 blocks** with pole sums within ±0.5 %, strong and weak poles alternating.

M10 Glue one pole: 4 blocks of the same part, with the jig NP1-T02 (Rev D shoe)

Drum upright, open (rear) end up, on the non-magnetic turntable. Comb bars and front end ring already cured.



Bonding is as doc 18 §6.4 and §6.5 with these changes:

1. Comb bars NP1-221 are trapezoid (wide face on the seat), **two per pole gap** (front row and rear row), and their outer ends sit in the notches of the end rings NP1-222; locate notch 1 from hub bolt hole 1. The indexing template has two positions per gap: front −0.25°, rear +0.25° from the nominal pole centre line (0.50 ±0.02° apart). The rear ring is clocked to the rear bars.
2. The jig NP1-T02 gets a **Rev D shoe**: a flat POM pocket 20.10 × 40.10 with the block held dot up (N poles) or dot down (S poles).
3. Per pole: block 1 against the left front comb bar, block 2 next to it (it moves against the right bar by itself), PTFE shim 0.2, then blocks 3 and 4 the same way against the rear bars (0.50° turned), clamp bar on all four until cured. The blocks start 3.1 mm before the stack (x 339.9), on the front end ring.
4. N poles first (all 20, dot up), cure, then S poles (dot down); gaussmeter check of every pole before the glue gels.
5. Pole map (doc 18 §6.6) with the pole centre limit tightened to **±0.05°** (§7), measured on each row separately; the rear row must read 0.50 ±0.02° after the front row on average over the 40 poles. The axial uniformity check looks for a dip over the single row joint at x 380.
6. Comb bars are placed with the indexing template to **±0.03°** (±0.16 mm).

10. Changes to other documents

Document	Change
Doc 14 §4.5	Magnets are 160 flat blocks NP1-220 Rev D, 40 × 20 × 7.50 (2 × 2 per pole, rows skewed 0.50°, 3.1 mm overhang each end); segmentation confirmed
Doc 14 §5.2	Remove “magnet thickness and bond line 0.050” from the air gap budget (mandrel-moulded sleeve)
Doc 14 open point 3	Closed by §3
Doc 18 §0 table, App. B	Magnets: 160 + 8 flat blocks 45.6 g, 7.30 kg; seat Ø618.65 × 96; EMF ≈ 380 V at 100 °C (2D, plus 0 to 2 % from the overhang), worst-case peak ≈ 895 to 910 V (28 turns)
Doc 18 M7, M8	Closed (one part number; eddy loss done)
Doc 18 §4.2	Comb bars: G11 trapezoid 7.04/7.67 × 7.0 × 44, 80 + 8 (NP1-221); end rings NP1-222 7.9 wide with notches, rear ring clocked 0.50°
Doc 18 §4.3	Replace by §8 of this document
Doc 18 §6.3, drum NP1-210	Magnet seat **Ø618.65 +0.05/0, x 332 to 428** (was Ø618.4, x 335 to 425); the thick shell band (OD Ø646) grows by 3 mm at each end to match
Doc 18 §6.7	Sleeve mould length x 332 to 428 (covers the overhanging blocks and both end rings)
Doc 18 §6.4, §6.5	As §9 of this document; comb bar placement ±0.03° (was ±0.05°)
Doc 18 §6.6	Pole centre acceptance ±0.05° (was ±0.1°)
Doc 06	Rotor loss 40 W confirmed (magnet eddy 15.5 W at rated)
`cad/nacelle_p1/geometry.py`	`MAGNET` still describes the doc 18 arcs; switch to `magnet_geom.py` values when doc 18 goes to Rev B (after Q1)

11. Open items

#	Item	Owner	Due
1	Approve this magnet definition (Rev B: 40 × 20 × 7.50 blocks, rows skewed 0.50°)	Martin	Now
2	Send §8 and drawing M03 for quotation (N40EH and N40UH) to at least three makers, at least one outside China; first article 5 pieces	Cat. 10	Week 1
3	Order 8 test pieces early for the Q1 magnet panels (unmagnetised acceptable)	Cat. 10	Week 1
4	Drawings NP1-221 comb bar, NP1-222 end ring, jig shoe NP1-T02 Rev D, indexing template with the two skew positions; drum drawing NP1-210 seat Ø618.65 × 96	Cat. 4/10	Week 2
4a	Measure the no-load EMF and cogging on Q1/P1 and compare with §13 (overhang and skew are 2D estimates); turn count 27 or 28 is the lever if the EMF comes out high	Cat. 11	Bench T1
5	Update doc 18 to Rev B with §10 (after Q1)	Cat. 10	Weeks 3 to 8
6	Bench check of magnet temperature at rated load (confirms §3)	Cat. 11	Bench T2

12. Magnet grade assessment

Question (Martin, 2026-11-05): which grade is best for this generator, which is easiest to buy, and which is feasible?

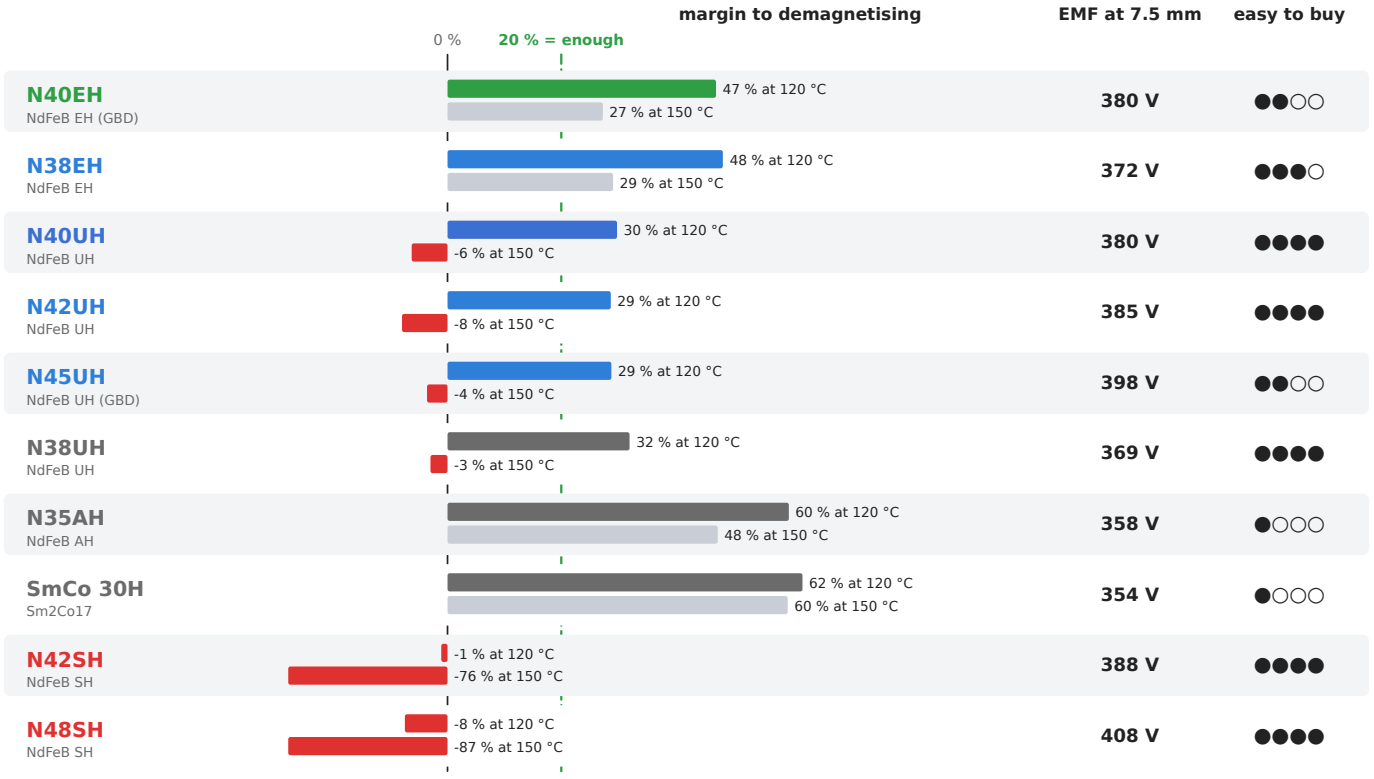
12.1 What the grade must do

The grade sets two things on the fixed block size: the **EMF** (through the remanence Br) and the **resistance to demagnetising** in the worst fault (through the intrinsic coercivity HcJ, which falls as the magnet warms). The design case is the doc 06 transient short circuit with the magnets at **120 °C**; a margin of at least **20 %** between the worst field in the magnet and HcJ is required. 150 °C is run as an extra check, because the magnet temperature in service is still a model value until the bench test.

Every grade was run on the final geometry with its catalogue minimum values (`grades.py` → `grades_summary_C2t.json` for Rev B; Rev A in `grades_summary.json`). EMF values include the 0.4 % skew loss.

M11 Which magnet grade? Safety against demagnetising, EMF, and how easy it is to buy

Same 40 × 20 × 7.5 block for every grade. Worst short circuit with the magnets at 120 °C (design case) and 150 °C (extra).



Ask every maker for N40EH AND N40UH on the same drawing. Same size, same EMF (380 V):
N40EH is the better magnet (safe even at 150 °C), N40UH is the easier one to buy and still meets the 120 °C design case.
SH grades fail the short-circuit check. AH and samarium-cobalt are safe but weak: blocks would need 9.7 to 10.3 mm.

Grade	Br / HcJ at 20 °C (min)	Line EMF, 7.50 mm block	Block for 380 V	Margin at 120 °C	Margin at 150 °C	Heavy rare earth (Dy/Tb)
N42SH	1.29 T / 1,592 kA/m	388 V	6.8 mm	**−1 %** (fails)	−76 %	low, ≈ 2 to 4 %
N48SH	1.37 T / 1,592 kA/m	408 V	5.1 mm	**−8 %** (fails)	−87 %	low

Grade	Br / HcJ at 20 °C (min)	Line EMF, 7.50 mm block	Block for 380 V	Margin at 120 °C	Margin at 150 °C	Heavy rare earth (Dy/Tb)
N38UH	1.22 T / 1,990 kA/m	369 V	8.6 mm	32 %	−3 %	≈ 4 to 6 %
N40UH	1.26 T / 1,990 kA/m	**380 V**	7.5 mm	**30 %**	−6 %	≈ 4 to 6 % (≈ 1 % with GBD)
N42UH	1.28 T / 1,990 kA/m	385 V	7.0 mm	29 %	−8 %	≈ 4 to 6 %
N45UH (GBD)	1.32 T / 1,990 kA/m	399 V	5.8 mm	29 %	−4 %	≈ 1 to 2 % Tb
N38EH	1.22 T / 2,388 kA/m	372 V	8.3 mm	48 %	29 %	≈ 6 to 9 %
N40EH (GBD)	1.25 T / 2,388 kA/m	**380 V**	7.5 mm	**47 %**	**27 %**	≈ 1.5 to 3 % Tb
N35AH	1.17 T / 2,786 kA/m	358 V	9.7 mm	60 %	48 %	≈ 9 to 12 %
SmCo 30H (Sm2Co17)	1.08 T / 1,990 kA/m	354 V	10.3 mm	62 %	60 %	none (samarium and cobalt instead)

Rev B margins are 1 to 3 points lower than Rev A’s (thinner block, wider pole), and the ranking is unchanged.

“Block for 380 V” is an estimate from the EMF slope found in §2; anything other than about 7.5 mm means new block, seat and sleeve numbers. Heavy rare earth contents are typical industry values, not supplier data.

12.2 Sourcing (status 2026-11-05)

- **Every grade that passes contains dysprosium or terbium** (or, for SmCo, samarium). China’s MOFCOM Announcement No. 18 (4 April 2025) puts seven medium and heavy rare earths, including Dy, Tb and Sm, **and magnets containing them** under export licensing. It has never been suspended: each shipment needs a licence, unless the maker holds one of the year-long general licences issued since December 2025 (first to JL MAG, Zhong Ke San Huan and Yunsheng). Dy and Tb exports are reported at about half their earlier level.
- A wider second wave of controls announced in October 2025 was suspended, and **that suspension runs out on 10 November 2026**, five days after this document. What follows is not known yet, so quotes should state the lead time including any export licence.
- **Makers outside China** produce UH and EH grades: in Japan (Shin-Etsu, Proterial, TDK), in Germany (VAC) and, since 2025, in Estonia (Neo Performance Materials, Narva). They are usually dearer and prefer larger orders; 7 kg per generator is small, so a distributor may be needed. Their plants also depend partly on Chinese heavy rare earth feed.
- **UH grades are the most common high-temperature grades**: wind, servo and traction motors use them, many makers stock the alloy, and N40UH and N42UH are standard. **EH grades** are made by fewer makers; N40EH needs grain-boundary diffusion (GBD) of terbium, which only the larger makers run. **AH and SmCo** are specialist products with few makers, long lead times and the highest prices.

12.3 Ranking

Rank	Grade	Best for the turbine	Easy to buy	Feasible for P1
1	**N40EH (GBD)**	★★★★ Same size and EMF as N40UH, safe even at 150 °C (27 %)	★★ Larger makers only, Tb content	**Yes: drop-in on the same drawing, ask for it**
2	**N40UH**	★★★ Meets the 120 °C case with 30 %, no reserve at 150 °C	★★★★ Standard grade, many makers	**Yes: current choice, safe fallback**
3	N38EH	★★★ Like N40EH but 2 % less EMF (block 8.3 mm or 29 turns)	★★★ Common EH grade	Yes, with a geometry change
4	N42UH	★★★ +1.4 % EMF over N40UH, same margin	★★★★	Yes, drop-in (EMF slightly above target)
5	N45UH (GBD)	★★ High EMF, but no reserve at 150 °C	★★	Only with a thinner block (5.8 mm)
6	N38UH	★★ Low EMF	★★★★	Needs 8.6 mm block
7	N35AH	★★ Very safe, low EMF	★	Needs 9.7 mm; long lead time
8	SmCo 30H	★★ Most temperature-proof, low EMF, brittle	★	Needs 10.3 mm and a new drum seat; costly
–	N42SH, N48SH	✗ Demagnetise in the worst short circuit	★★★★	**No** (as doc 06 found for N42SH)

Recommendation. Keep the drawing as it is (40 × 20 × 7.50) and ask every maker to quote **N40EH (preferred) and N40UH (accepted)** on the same drawing. Both give 380 V. N40EH gives 27 % margin even at 150 °C, which covers an error in the magnet temperature model at no cost in size or EMF. N40UH meets the 120 °C design case with 30 % margin and is the easier part to buy. Choose N40EH if a qualified maker can deliver it, with its export licence, inside the 8 to 12 weeks planned in doc 18; otherwise N40UH. The bench test (doc 15) measures the real magnet temperature and settles which grade the production version needs.

13. Rev B: 40 × 20 mm catalogue blocks

Question (Martin, 2026-11-05): adjust the design for 40 × 20 magnets.

13.1 Two ways to lay a 40 × 20 block

“40 × 20” is taken as the face size of the block; the thickness (the magnetised direction) stays free, because makers grind it to order from stock blocks. The pole pitch at the magnet face is 47.4 mm and the stack 74 mm, so the block can lie two ways:

	C2 / C2t: 20 across, 40 along	C1: 40 across, 20 along
Blocks per pole	2 across × 2 along	1 across × 4 along
Pole span	7.64° (0.85)	7.59° (0.84)
Length of the rows	40 + 0.2 + 40 = 80.2 mm	4 × 20 + 3 × 0.2 = 80.6 mm
Bond under the block middle	0.31 mm	0.80 mm (wide flat block on a round seat)
EMF at 8.2 mm thick	389 V	375 V
Cogging without / with skew	6.9 / < 0.1 Nm (2 rows, 0.50°)	2.7 / < 0.1 Nm (4 rows, 0.25° steps)
Magnet eddy loss	14.8 W	10.0 W
Demag margin, N40UH, 120 °C	31.2 %	28.7 %

20 across, 40 along wins: the wide single block (C1) wastes magnet (the edges stand far off the stator, like F1 in §2), needs a thicker and heavier block for the same EMF, has a 0.8 mm glue line under its middle, three shim joints and four skew steps. Its only gain is 5 W less eddy loss.

13.2 Thickness

With 20 across, the 0.85 span gives more flux than Rev A’s 0.80, so the block can be **thinner: 7.50 mm gives 381 V before skew, 380 V after** (28 turns, 100 °C), the doc 06 target. 8.2 mm would give 389 V and 7.0 mm 375 V. The seat follows the thickness: Ø618.65 for 7.50.

If a maker offers a stock thickness instead of grinding 7.50: **8.0 mm** gives about 385 V and Ø619.6 (acceptable with 28 turns, peak voltage ≈ 905 V); **10 mm** gives an estimated 400 V and Ø623.3, which would need 27 turns and new drum, sleeve and comb numbers. Ask for 7.50.

13.3 The overhang

Two 40 mm blocks and the 0.2 mm shim make 80.2 mm against the 74 mm stack. The rows are centred on the stack, so the magnets **overhang 3.1 mm at each end**. This is normal practice in PM machines and does no harm: the overhanging magnet sits on the same seat, under the same sleeve, and the end turns are well inside it (r < 299). It adds a little end flux, typically 1 to 2 % EMF for this gap, which the 2D FEA does not include; the EMF is therefore 380 to 387 V and is measured on Q1 (§11 item 4a). The alternative, grinding the blocks to 36.9 mm, would give up the catalogue size, which was the point of the change.

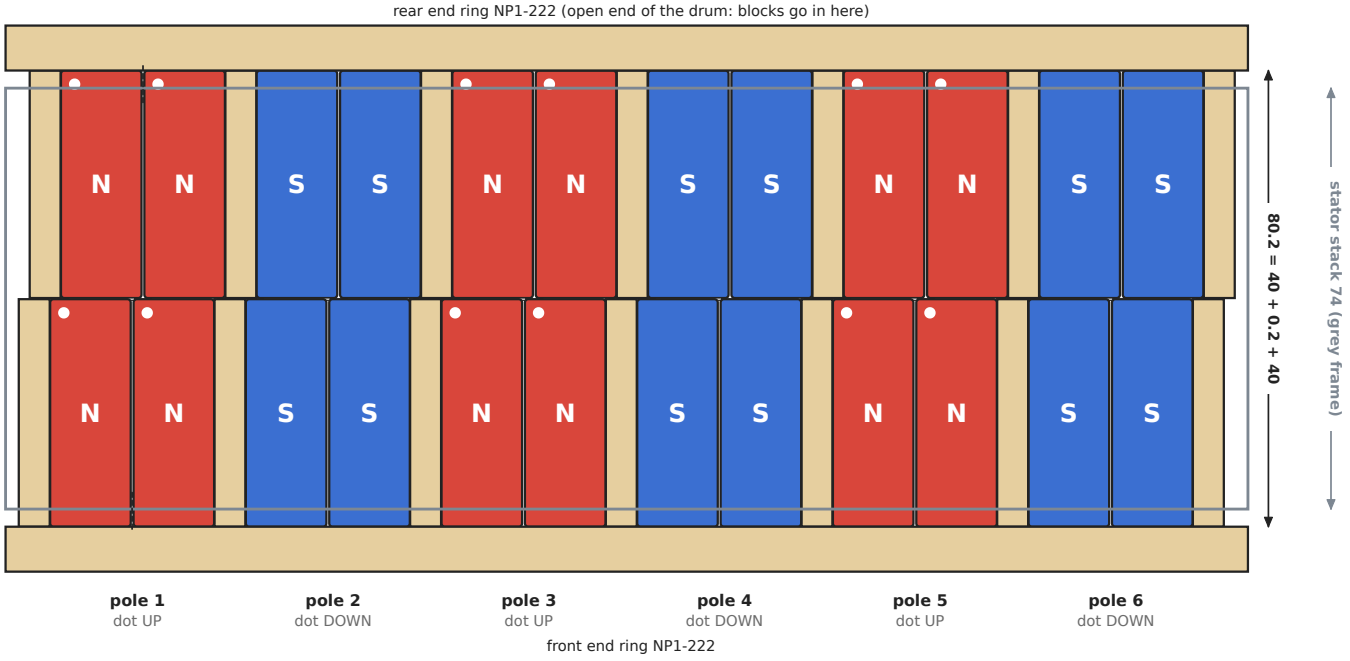
The drum seat land grows from x 335–425 to **x 332–428** so the 7.9 mm end rings still fit. If a drum were already machined, 4.9 mm end rings on the old land would also work; the drum is not made yet, so the land is lengthened instead.

13.4 Why the rows are skewed

The 40-pole, 45-slot machine has a cogging period of 1° (360 / lcm(40, 45)). Its size depends strongly on the pole span: 0.80 (Rev A) sits in a valley (≈ 1 Nm), 0.84 to 0.85 on a peak (5.4 to 6.9 Nm, §2). The 20 mm block width fixes the span at 0.85. Turning the rear row by **half the cogging period, 0.50°**, makes the two rows cog in opposite phase, so their sum cancels. In 2D (the two rows superposed) the cogging falls from 6.6 to 0.02 Nm and the load ripple from 2.83 to 1.97 %; the EMF loses only 0.4 % (the fundamental sees a 10° electrical skew). End effects at the row joint are not in the model, so the bench measures the real cogging (§11 item 4a).

M06 Where every block goes: the drum seat unrolled, seen from the stator

Part of the ring, 6 of the 40 poles. Each pole = 2 blocks across × 2 blocks along = 4 blocks of the SAME part.



tan = G11 comb bars and end rings
white strip = 0.2 mm PTFE shim (taken out after cure)

- rear row turned 0.50° (2.7 mm) against the front row
- blocks overhang the stack 3.1 mm at each end
- white dot visible = N pole (dot faces the stator)
- N and S always alternate, 20 + 20 poles

The skew costs nothing in parts: the comb bars are split into a front and a rear bar per gap anyway (each 44 mm long), so the rear set is simply placed 0.50° further round with the indexing template. The blocks, end rings and jig are the same for both rows.

Appendix A. How to rerun

```
cd analysis/generator
python3 magnets.py              # all 11 shapes, ~70 min (or in parallel: A F1 F2 F3 / F2w F2t
F2wt / C2 C1 C2t C2s)
python3 magnets.py tol F2t      # tolerance scatter on the Rev A shape, ~20 min
python3 grades.py               # magnet grade assessment on Rev A (F2t), ~10 min
python3 magnets.py C2 C1 C2t C2s   # Rev B: 40 x 20 cases incl. step skew, ~30 min
python3 magnets.py tol C2t      # Rev B tolerance scatter, with and without skew
python3 grades.py C2t           # Rev B grade assessment (§12) -> grades_summary_C2t.json
cd ../../cad/generator_mfg
python3 magnet_geom.py          # final dimensions
python3 figures_magnets.py --png
```